



Lesson 1.4 — Function Notation and Evaluation

Big idea: $f(x)$ is not multiplication. It's a label that says "the output of function f when the input is x ." Get comfortable swapping the input, and everything else in Algebra 2 gets easier.

Quick review

Read it: $f(3)$ means "the output when the input is 3." Not "f times 3."

Evaluate at a number: substitute the number for every x in the rule and simplify.

Evaluate at an expression: substitute the entire expression (with parentheses) for every x .

Read from a graph: $f(a) = b$ means the graph of f passes through the point (a, b) .

Read from a table: find the row where the input equals a ; the output in that row is $f(a)$.

Worked example

Worked example. Let $f(x) = 2x^2 - 5$. Find $f(3)$, $f(-1)$, and $f(x + 1)$.

$$f(3) = 2(3)^2 - 5 = 13. \quad f(-1) = 2(-1)^2 - 5 = -3. \quad f(x + 1) = 2(x + 1)^2 - 5 = 2x^2 + 4x - 3.$$

Warm-ups

1. Let $f(x) = 3x - 7$. Find $f(4)$, $f(0)$, and $f(-2)$.
2. Let $g(x) = x^2 + 1$. Find $g(2)$, $g(-3)$, and $g(0)$.
3. Let $h(x) = \sqrt{x + 4}$. Find $h(5)$, $h(0)$, and $h(-4)$.

Practice

1. Let $f(x) = 2x + 1$. Find $f(a)$, $f(a + 3)$, and $f(2a)$.
2. Let $g(x) = x^2 - 4x$. Find $g(x + 2)$ and simplify.
3. From the table, find $f(1)$, $f(-2)$, and the x -value where $f(x) = 7$.
x: -3, -2, -1, 0, 1, 2, 3
f(x): 16, 9, 4, 1, 0, 1, 4

Challenge

1. Let $f(x) = x^2$. Simplify the expression $[f(x + h) - f(x)] / h$, assuming $h \neq 0$. (This is the setup for the derivative in calculus — get comfortable with it now.)
2. The function $P(t)$ gives the population of a town (in thousands) t years after 2020. Interpret each in plain English: (a) $P(0) = 42$, (b) $P(5) = 58$, (c) $P(10) - P(5) = 7$.



Answer key — Lesson 1.4

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $f(4) = 5$, $f(0) = -7$, $f(-2) = -13$.
2. $g(2) = 5$, $g(-3) = 10$, $g(0) = 1$.
3. $h(5) = 3$, $h(0) = 2$, $h(-4) = 0$.

Practice

1. $f(a) = 2a + 1$, $f(a + 3) = 2a + 7$, $f(2a) = 4a + 1$.
2. $g(x + 2) = (x + 2)^2 - 4(x + 2) = x^2 - 4$.
3. $f(1) = 0$, $f(-2) = 9$. $f(x) = 7$ is not exactly in the table; between $x = -2$ and $x = -1$ the output drops from 9 to 4, so $f(x) = 7$ for an x -value near -1.7 (no exact table entry).

Challenge

1. $[(x + h)^2 - x^2] / h = [x^2 + 2xh + h^2 - x^2] / h = [2xh + h^2] / h = 2x + h$.
2. (a) In 2020, the population was 42,000. (b) In 2025, it was 58,000. (c) Between 2025 and 2030, the population grew by 7,000.